

7.04 Network Algorithms

7.04a	Least weight path between two vertices	a) Be able to use examples to demonstrate understanding and use of Dijkstra's algorithm to find the length and route of a least weight (shortest) path. <i>Solve problems that require a least weight (shortest) path as part of their solution.</i> <i>Know that Dijkstra's algorithm has quadratic order (as a function of the number of vertices).</i>	
7.04b	Least weight set of arcs connecting all vertices	b) Be able to use examples to demonstrate understanding and use of Prim's algorithm (both in graphical and tabular/matrix form) and Kruskal's algorithm to find a minimum connector (minimum spanning tree) for a network. <i>Solve problems that require a minimum spanning tree as part of their solution.</i> <i>Includes adapting a solution to deal with practical issues.</i> <i>Know that Prim's algorithm and Kruskal's algorithm have cubic order (as a function of the number of vertices).</i>	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
7.04c	Least weight cycle through all vertices		c) Be able to use the nearest neighbour method on a network formed by weighting a complete graph to find an upper bound for the travelling salesperson problem. <i>Includes understanding that when the nearest neighbour method is used on a network formed by weighting a graph that is not complete it may stall before reaching every vertex or it may reach every vertex but to close the route it may need a path that is not a direct connection from the end vertex back to the start vertex.</i> <i>Includes choosing between two, or more, upper bounds to find the best (least) upper bound.</i> <i>Includes using short-cuts where possible to improve an upper bound.</i>
7.04d			d) Be able to use a minimum spanning tree on a reduced network to calculate a lower bound for the travelling salesperson problem on a complete graph and understand why this method gives a lower bound. <i>Understand that for a graph that is not complete this method can give a value that is not a lower bound.</i> <i>Includes choosing between two or more lower bounds to find the best (greatest) lower bound.</i>

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
7.04e	Least weight route through all vertices that traverses every arc at least once		e) Be able to solve the route inspection problem by consideration of all possible pairings of up to six odd nodes. <i>If a problem has more than six odd nodes additional restrictions will reduce the number of pairings that need to be considered.</i> <i>Problems may be set that require an understanding of how the number of pairings increases as the number of odd nodes increases. For $2n$ odd nodes the number of pairings is</i> $1 \times 3 \times \dots \times (2n - 1) = \prod_{r=1}^n (2r - 1).$
7.04f	Network problems	f) Be able to choose an appropriate algorithm to solve a practical problem. <i>Includes adapting an algorithm or a solution to deal with practical issues.</i>	